Mean-field Bayesian circuit learning at every width above target complexity

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Abstract

We prove a sample bound for Bayesian neural-network learning that depends on the size of a small target implementation and is independent of the learner's potentially much larger width. The model is a fixed-depth tanh network at mean-field Gaussian scaling: internal weight variance is $1/N$ and readout variance is $1/N^2$ at hidden width N. Before training, the prior retains only networks whose outputs have absolute value greater than $1/2$ on the entire Boolean input cube. Learning then conditions on noiseless labels. The sample bound is polynomial in input dimension and the target's bounded-weight network complexity, uniformly at every width at least that complexity. For any common training and test input distribution, sufficiently many examples make the posterior assign the correct label probability strictly greater than $2/3$ on almost all test inputs. The proof bounds the total prior probability of functions lacking a small description and compares it with the probability assigned to the target. Its main step compresses likely hidden-layer covariance patterns into finite networks. All arguments apply at finite width; no infinite-width limit is required. The polynomial degree and constants depend on depth, so the theorem does not provide a bound polynomial in depth.

1 Introduction

A network with many more parameters than training examples can represent many functions that fit the data. To explain why Bayesian learning should nevertheless generalize, one would like a sample bound determined by the complexity of the target, rather than by the number of parameters available to the learner. This paper proves such a bound for a Gaussian neural-network prior at mean-field scaling.

The model is a fully connected tanh network of depth L and hidden width N, with a linear readout. The Gaussian variances are $1/d$ in the input layer, $1/N$ in internal layers, and $1/N^2$ in the readout. We retain only networks with absolute output margin greater than $1/2$ at every Boolean input, then condition on exact agreement with noiseless training labels. This defines a probability distribution over Boolean functions and, after training, a posterior distribution over their predictions. The global margin restriction is part of the prior, including at unobserved inputs.

Suppose the target has a depth-L, width-C implementation with that strict margin and all weights bounded by C. Our result gives a sample bound polynomial in d, C, the inverse desired error, and the logarithm of the inverse failure probability, uniformly for every learner width $N \geq C$. It holds for any common training and test distribution on the Boolean cube. In particular, on almost all test inputs the posterior gives the correct label probability greater than $2/3$. Section 2 states the precise guarantee in terms of the smallest such implementation size $C_L(f)$.

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The result is finite-dimensional throughout. It does not require the learner to be much wider than the target, and it does not take a width limit before training. The complexity measure is a bounded-weight network complexity at the specified depth, so no translation from arbitrary Boolean circuits is implicit. Depth remains fixed in the sample bound: its polynomial degree and constants may grow rapidly with L. As with an Occam learning bound, the conclusion is statistical and does not establish an efficient method for computing the posterior.

How excess width is controlled. We compare the prior mass of the target with the total mass of functions that cannot be described by small networks. A small target can be embedded at any larger width by replicating its neurons; a neighborhood of this embedding supplies a lower bound on its prior mass. For the competing functions, we show that all but a sufficiently small amount of prior mass can be compressed into one finite family whose size is independent of N. Training labels rule out the inaccurate members of this family by an elementary union bound. This last step explains why a bound on the *total* exceptional mass is needed, rather than a separate small probability estimate for each complicated function.

The main difficulty is to construct that finite family at the mean-field scaling, where the random internal matrices retain operator norm of constant order as width grows. We work instead with the correlations of hidden activations across inputs. Finite exponential tests, a convex duality argument, and covariance stability permit these correlations to be represented by smaller networks. The formal proof below supplies all three ingredients and then completes the posterior estimate.

2 Model and main result

Let $d \geq 1$, $L \geq 2$, and $\mathcal{X} = \{-1, 1\}^d$. The augmented input is $\bar{x} = (1, x_1, \ldots, x_d)$. A depth-$L$ network has $L - 1$ hidden layers of common width N and computes

$$h^0(x) = \bar{x}, \qquad h^s(x) = \tanh(W^s h^{s-1}(x)) \quad (1 \leq s < L), \qquad z_w(x) = W^L h^{L-1}(x).$$

There are no additional biases. The product Gaussian prior μ_N has independent centered entries with standard deviations

$$\sigma_1 = d^{-1/2}, \qquad \sigma_s = N^{-1/2} \quad (2 \leq s < L), \qquad \sigma_L = N^{-1}.$$

These are the mean-field conventions used here. In the broader family with variances $N^{1-\gamma}/d$, $N^{-\gamma}$, and $N^{-\gamma-1}$, they correspond to $\gamma = 1$; no other scaling regime is needed in this paper. For a Boolean function g write

$$A_{N,g} = \{w : g(x) z_w(x) > 1/2 \text{ for every } x \in \mathcal{X}\}, \qquad A_N = \bigcup_g A_{N,g}, \qquad \pi_N(g) = \frac{\mu_N(A_{N,g})}{\mu_N(A_N)}.$$

Thus the global margin restriction is imposed before the function prior is conditioned on training labels. Let $C_L(f)$ be the least integer $C \geq 1$ for which a depth-L, width-C network implements f with this strict margin and every weight bounded in absolute value by C. We assume $C_L(f) < \infty$.

For independent inputs $x_1, \ldots, x_n \sim D$, the training sample is $S_n = ((x_i, f(x_i)))_{i=1}^n$, and the posterior is

$$\Pi_N(g \mid S_n) = \frac{\pi_N(g)\, \mathbf{1}\{g(x_i) = f(x_i)\ \forall i\}}{\sum_h \pi_N(h)\, \mathbf{1}\{h(x_i) = f(x_i)\ \forall i\}}.$$

Its average error is $R_D(\Pi_N) = \mathbb{E}_{g \sim \Pi_N} \mathbb{P}_{x \sim D}\{g(x) \neq f(x)\}$.

Theorem 2.1 (Learning at every width above target complexity)**.** *For every fixed depth $L \geq 2$ there is a polynomial p_L such that the following holds. Let $C = C_L(f) < \infty$, let $N \geq C$, and let $\varepsilon, \delta \in (0,1)$. For every distribution D on $\mathcal{X}$, if*

$$n \geq \frac{6}{\varepsilon}\left[p_L\left(d, C, \log\frac{1}{\varepsilon}\right) + \log\frac{6}{\varepsilon\delta}\right], \tag{1}$$

then, with probability at least $1-\delta$ over S_n,

$$R_D(\Pi_N) \leq \varepsilon/3, \qquad \mathbb{P}_{x\sim D}\{\Pi_N(g(x) = f(x) \mid S_n) > 2/3\} \geq 1-\varepsilon.$$

The polynomial is independent of width. Its degree and constants may depend on depth. The proof gives explicit quantities that can replace p_L.

The statement is information-theoretic: efficient posterior computation or efficient construction of the compressed network is not asserted. A fixed confidence threshold greater than $1/2$ can replace $2/3$, with the corresponding change in constants.

Proof roadmap. Section 3 constructs a Gaussian neighborhood of the embedded target with probability at least $e^{-N\,\mathrm{poly}_L(d,C)}$. Section 4 develops finite tests for empirical hidden-layer covariances. These tests identify a family of controlled changes of measure, called quadratic tilts, whose covariances approximate the observed ones; the exceptional probability is exponentially small in N. Section 5 proves stability of the tilted covariances and represents them by finite networks. We then choose the approximation budgets backward through the layers, obtaining a description bound independent of learner width. The final section compares the exceptional mass with the target mass and applies a finite-class posterior estimate. A direct target-mass estimate covers every smaller width where the compression probability bound is unavailable.

3 A target-mass bound at every admissible width

The following construction bounds the target's prior cost at every width $N \geq C$, at the mean-field exponent.

Proposition 3.1 (Uniform cloning mass bound)**.** *Fix $L \geq 2$ and an activation ϕ satisfying $|\phi| \leq 1$ and having Lipschitz constant at most one. There is a function $Q_L(d, C)$ bounded by a polynomial such that for every Boolean target with $C_L(f) \leq C$, and every $N \geq C$,*

$$-\frac{2}{N}\log \mathbb{P}\{f(x)z_w(x) > 1/2 \text{ for all } x\} \leq Q_L(d, C).$$

The prior variances and absence of later-layer biases are exactly those in the main model.

Proof. Choose a width-C teacher, increase its readout by a factor of three, and write its hidden matrices as B^s and readout as v. Thus the teacher has margin strictly greater than $3/2$, hidden entries bounded by C, and readout entries bounded by $3C$. If the original witness has smaller width, first use its actual width in the construction below; all estimates only improve. Partition $\{1, \ldots, N\}$ into C groups of sizes k_a differing by at most one. Then

$$1 \leq k_a, \qquad \frac{\max k_a}{\min k_a} \leq 2, \qquad \frac{N}{k_a} \leq 2C.$$

Let $u_a = k_a^{-1/2}\mathbf{1}_{\text{group } a}$ and let P be the orthogonal projection onto their span. The exact clone w_* has first-layer row B_b^1 for a neuron in group b, hidden entry B_{ba}^s/k_a from group a to group b, and readout v_b/k_b on group b. Every hidden activation of the exact clone is group-constant. Moreover,

$$\|W_*^s\|_{\text{op}} \le \sqrt{2}C^2 \quad (2 \le s < L), \qquad \sqrt{N}\|W_*^L\|_2 \le 3\sqrt{2}C^2.$$

The first bound follows by restricting to the group-constant subspace: the resulting matrix has entries $\sqrt{k_b/k_a}B_{ba}^s$, and its Frobenius norm is at most $\sqrt{2}C^2$.

Set $t = 1$. The random coordinates $\sqrt{d}W_{ij}^1$, $\sqrt{N}W_{ij}^s$ for $2 \le s < L$, and $\alpha_j = NW_j^L$ are independent centered Gaussians of standard deviation t. Impose the following cylinder event $\mathcal{C}_\rho$, for $0 < \rho \le 1$:

- every $\sqrt{d}W_{ij}^1$ is within ρt of $\sqrt{d}(W_*^1)_{ij}$;
- for every hidden row j and group a, the coordinate $\langle\sqrt{N}W_{j,\cdot}^s, u_a\rangle$ is within ρt of its cloned value $\sqrt{N/k_a}B_{ba}^s$;
- every α_j is within ρt of its cloned value Nv_b/k_b.

All these constraints concern independent scalar Gaussian coordinates. The orthogonal coordinates of each internal row remain independent prior Gaussians. Consequently write

$$W^s = W_*^s + R^s + E^s, \qquad R^s = R^sP, \quad E^sP = 0.$$

On the cylinder, $\|R^s\|_{\text{op}} \le \|R^s\|_F \le \rho t\sqrt{C}$. The free matrix is distributed as $(t/\sqrt{N})G^s(I-P)$, with G^s a standard Gaussian matrix, independently of the cylinder coordinates. Let

$$b_L = \sqrt{16\log 9 + 8\log(4L)}.$$

For completeness, a $1/4$-net of the unit sphere in $\mathbb{R}^N$ has at most 9^N points, by the usual disjoint-ball volume argument. For a matrix G, approximating both unit vectors in its bilinear form by this net gives $\|G\|_{\text{op}} \le 2\max_{u,v \text{ in the net}} |u^\mathsf{T}Gv|$. For standard Gaussian G, each displayed bilinear form is standard normal. Its elementary Chernoff bound and a union bound give

$$\mathbb{P}\{\|G\|_{\text{op}} > b_L\sqrt{N}\} \le 2 \cdot 9^{2N}e^{-b_L^2N/8} = 2(4L)^{-N} \le (2L)^{-1}.$$

Thus a union bound implies, conditionally on $\mathcal{C}_\rho$, that all $\|E^s\|_{\text{op}} \le b_L t$ with probability at least $1/2$. This is also valid when $L = 2$, where there are no such matrices.

Let $e_s(x) = N^{-1/2}\|h_w^s(x) - h_*^s(x)\|_2$. The input constraints give $e_1(x) \le \rho(d+1)/\sqrt{d} \le \rho(d+1)$. Since $E^sh_*^{s-1} = 0$, boundedness and Lipschitz continuity give, simultaneously for every input,

$$e_s(x) \le (\sqrt{2}C^2 + \sqrt{C} + b_L)e_{s-1}(x) + \rho\sqrt{C}.$$

Put $A = 2 + 3C^2 + b_L$. Induction yields $e_s(x) \le \rho(d + L + 1)A^{L-2}$ for $s < L$. At the output,

$$|z_w(x) - z_*(x)| \le 3\sqrt{2}C^2e_{L-1}(x) + \rho.$$

Hence choosing

$$\rho = \left[2 + 9C^2(d + L + 1)A^{L-2}\right]^{-1}$$

ensures output discrepancy at most $1/2$. All networks in the cylinder with the operator-norm event therefore implement f with margin greater than one.

It remains to price the cylinder. Its number of constrained coordinates is

$$m = N\big[d + 2 + (L-2)C\big].$$

The sum of their squared centers is at most NB_0, where one may take

$$B_0 = d(d+1)C^2 + 2(L-2)C^4 + 36C^4.$$

Indeed, a first-layer center has magnitude at most $\sqrt{d}C$, an internal center at most $\sqrt{2C}C$, and a readout center at most $6C^2$. For $Z \sim N(0, t^2)$ and $|Z - a| \le \rho t$,

$$\mathbb{P}\{|Z-a| \le \rho t\} \ge \frac{2\rho}{\sqrt{2\pi}} \exp\{-a^2/t^2 - \rho^2\}.$$

Multiplication and $t = 1$ show

$$-\log \mathbb{P}(\mathcal{C}_\rho) \le NB_0 + N[d + 2 + (L-2)C] \log \frac{e\sqrt{2\pi}}{2\rho}.$$

The desired probability is at least $\mathbb{P}(\mathcal{C}_\rho)/2$. Since $N \ge 1$, this proves the claim with

$$Q_L(d, C) = 2B_0 + \frac{4[d + 2 + (L-2)C]}{\rho} + 2.$$

Here we used $\log(e\sqrt{2\pi}/(2\rho)) \le 2/\rho$ for $0 < \rho \le 1$. The displayed expression is a polynomial in d, C at every fixed depth. □

4 A finite family of entropy certificates

The following bound replaces a discretization of the full feature space by sparse linear tests. Its logarithmic dependence on the number of observables is essential when those observables are indexed by all pairs of Boolean inputs.

Let $(\Omega, \mathcal{A}, \mu)$ be any probability space, let $F = (F_1, \dots, F_D) : \Omega \to [-1, 1]^D$ be measurable, and write

$$\Lambda(\lambda) = \log \int e^{\lambda \cdot F}\, d\mu, \qquad \Psi_G(q) = \max_{\|\lambda\|_1 \le G} \{\lambda \cdot q - \Lambda(\lambda)\}, \qquad G > 0.$$

The maximum exists: its domain is compact and its objective is continuous. The norm $\|\lambda\|_1$ here is the sum of the absolute values of its D coordinates.

Lemma 4.1 (A bounded exponential tilt with controlled error)**.** *For every $q \in \mathbb{R}^D$, there is a maximizer λ_* in the preceding display such that the probability measure*

$$\frac{d\nu_*}{d\mu} = \exp\{\lambda_* \cdot F - \Lambda(\lambda_*)\}$$

satisfies the exact identity

$$\Psi_G(q) = D(\nu_* \| \mu) + G \left\| q - \int F\, d\nu_* \right\|_\infty . \tag{2}$$

Consequently, if $\Psi_G(q) \le K$, then

$$D(\nu_* \| \mu) \le K, \qquad \left\| q - \int F\, d\nu_* \right\|_\infty \le K/G.$$

Proof. Boundedness of F permits differentiation under the integral and gives $\nabla\Lambda(\lambda_*) = \int F\,d\nu_* =: m_*$. The first-order condition for a differentiable concave function maximized over the convex ℓ^1 ball is

$$(q - m_*) \cdot (\lambda - \lambda_*) \le 0 \qquad (\|\lambda\|_1 \le G).$$

Taking the maximum of the left linear functional over that ball gives

$$\lambda_* \cdot (q - m_*) = G\|q - m_*\|_\infty.$$

Moreover, $D(\nu_*\|\mu) = \lambda_* \cdot m_* - \Lambda(\lambda_*)$. Adding these two identities proves (2). Relative entropy is nonnegative, which proves the stated consequences. The log density of ν_* relative to μ is bounded, so its entropy is finite. No support, density, or nonsingularity assumption on μ was used. □

For an integer $s \ge 1$, introduce the finite set

$$\mathcal{V}_{G,s} = \left\{ \frac{1}{s} \sum_{j=1}^{s} a_j : a_j \in \{0, Ge_1, -Ge_1, \ldots, Ge_D, -Ge_D\} \right\}.$$

Thus $\#\mathcal{V}_{G,s} \le (2D+1)^s$ and every vector in this set has ℓ^1 norm at most G.

Lemma 4.2 (Sparse approximation of the dual value)**.** *For every $q \in \mathbb{R}^D$,*

$$\Psi_G(q) \le \frac{G^2}{2s} + \max_{v \in \mathcal{V}_{G,s}} \{v \cdot q - \Lambda(v)\}. \tag{3}$$

Proof. Fix λ with $\|\lambda\|_1 \le G$. Let the random vector A take the value $G\,\mathrm{sign}(\lambda_i)e_i$ with probability $|\lambda_i|/G$ for each nonzero coordinate, and take value zero with the remaining probability. Then $\mathbb{E}A = \lambda$. Take independent copies $A_1, \ldots, A_s$ and put $V = s^{-1}\sum_j A_j \in \mathcal{V}_{G,s}$. For each fixed $\omega \in \Omega$, the scalar $A \cdot F(\omega)$ lies in $[-G, G]$. Hoeffding's exponential-moment inequality therefore gives

$$\mathbb{E}\exp\{(V - \lambda) \cdot F(\omega)\} \le \exp\{G^2/(2s)\}.$$

Tonelli's theorem and Jensen's inequality imply

$$\begin{aligned} \mathbb{E}\Lambda(V) &= \mathbb{E}\log \int e^{V \cdot F}\,d\mu \\ &\le \log \mathbb{E} \int e^{V \cdot F}\,d\mu \\ &\le \Lambda(\lambda) + G^2/(2s). \end{aligned}$$

Since $\mathbb{E}(V \cdot q) = \lambda \cdot q$, at least one realization $v \in \mathcal{V}_{G,s}$ satisfies

$$v \cdot q - \Lambda(v) \ge \lambda \cdot q - \Lambda(\lambda) - G^2/(2s).$$

Take λ to be a maximizer defining $\Psi_G(q)$. □

Proposition 4.3 (A nonasymptotic entropy certificate)**.** *Let $X_1, \ldots, X_N$ be independent with law μ, and put $q_N = N^{-1}\sum_{i=1}^N F(X_i)$. For every $K > 0$, $G > 0$, and integer $s \ge 1$,*

$$\mathbb{P}\{\Psi_G(q_N) > K\} \le \exp\left\{ s\log(2D+1) - N\left(K - \frac{G^2}{2s}\right)\right\}. \tag{4}$$

Outside this exceptional event, the empirical moment vector is within K/G in maximum norm of the moments of an exponential tilt with parameter ℓ^1 norm at most G and relative entropy at most K. In particular, choosing $s = \lceil G^2 \rceil$ bounds the right side by $\exp\{\lceil G^2 \rceil \log(2D+1) - N(K - 1/2)\}$.

Proof. For each fixed $v \in \mathcal{V}_{G,s}$, independence gives exactly

$$\mathbb{E}\exp\{N[v \cdot q_N - \Lambda(v)]\} = 1.$$

Markov's inequality bounds the probability that its bracket exceeds $K - G^2/(2s)$ by $\exp\{-N(K - G^2/(2s))\}$. The sparse approximation lemma and a union bound over $\mathcal{V}_{G,s}$ prove (4). The bounded-dual lift lemma gives the final assertion. □

Application to hidden-layer covariances. Take m input points, $D = m^2$, $\mu = \mathcal{N}(0, Q)$ for any positive semidefinite covariance Q, and $F_{ij}(z) = \tanh(z_i)\tanh(z_j)$, indexed by all ordered pairs (i, j). Reshape a maximizing parameter λ_* into a matrix and replace it by its symmetric part Γ. This preserves its quadratic form and does not increase its entrywise ℓ^1 norm. The lifted covariance is then exactly

$$\mathsf{F}_\Gamma(Q) = \frac{\mathbb{E}_{Z\sim\mathcal{N}(0,Q)}[\tanh(Z)\tanh(Z)^\top e^{\tanh(Z)^\top \Gamma \tanh(Z)}]}{\mathbb{E}_{Z\sim\mathcal{N}(0,Q)} e^{\tanh(Z)^\top \Gamma \tanh(Z)}}.$$

Thus, except with the probability in (4), the empirical activation covariance $\widehat{Q}$ satisfies

$$\|\widehat{Q} - \mathsf{F}_\Gamma(Q)\|_{\max} \le K/G, \qquad \|\Gamma\|_{1,\mathrm{ent}} \le G.$$

This assertion is valid conditionally on any preceding network layers, using their empirical covariance as Q. Its probability bound is uniform in that covariance, including singular covariances. On the Boolean cube, $m = 2^d$ and the logarithmic factor is $\log(2m^2 + 1) \le (2d + 2)\log 2$, which is linear in d.

5 Covariance stability and finite realization

Write $\phi = \tanh$. Write $\|A\|_{\max} = \max_{ij}|A_{ij}|$ and $\|A\|_{1,\mathrm{ent}} = \sum_{ij}|A_{ij}|$ for matrices on the finite input set. These are different norms; their dual pairing is what avoids a factor of 2^d below. For a positive semidefinite covariance Q and a symmetric matrix Γ, let

$$\mathsf{F}_\Gamma(Q) = \frac{\mathbb{E}_{Z\sim\mathcal{N}(0,Q)}[\phi(Z)\phi(Z)^\top \exp\{\phi(Z)^\top \Gamma \phi(Z)\}]}{\mathbb{E}_{Z\sim\mathcal{N}(0,Q)}[\exp\{\phi(Z)^\top \Gamma \phi(Z)\}]}.$$

All activations are coordinatewise. The definition is valid for singular Q.

Lemma 5.1 (A dimension-independent covariance comparison)**.** *If $g = \|\Gamma\|_{1,\mathrm{ent}}$, then for all positive semidefinite Q, Q' of the same size,*

$$\|\mathsf{F}_\Gamma(Q) - \mathsf{F}_\Gamma(Q')\|_{\max} \le 8(1 + g)^2\|Q - Q'\|_{\max}.$$

Proof. Put $V(z) = \phi(z)^\top \Gamma \phi(z)$ and $Q_t = (1 - t)Q + tQ'$. For a twice differentiable bounded test function F, let $\mathbb{E}_t$ denote expectation under the Gaussian of covariance Q_t reweighted by e^V. Gaussian differentiation and the quotient rule give

$$\frac{d}{dt}\mathbb{E}_t F = \frac{1}{2}\sum_{a,b}(Q' - Q)_{ab}\{\mathbb{E}_t[\partial_{ab}F + \partial_a F \partial_b V + \partial_b F \partial_a V] + \mathrm{Cov}_t(F, \partial_{ab}V + \partial_a V \partial_b V)\}.$$

This formula has no reciprocal partition-function bound: all terms are expectations under a probability measure. Define $M_1(H) = \sup_z \sum_a |\partial_a H(z)|$ and $M_2(H) = \sup_z \sum_{a,b} |\partial_{ab} H(z)|$. Because $|\phi|, |\phi'| \le 1$ and $|\phi''| \le 2$,

$$M_1(V) \le 2g, \qquad M_2(V) \le 6g.$$

For $F(z) = \phi(z_i)\phi(z_j)$, including $i = j$, $\|F\|_\infty \le 1$, $M_1(F) \le 2$, and $M_2(F) \le 6$. The absolute derivative is therefore at most

$$\frac{1}{2}\|Q - Q'\|_{\max}\{6 + 8g + 2(6g + 4g^2)\} = (3 + 10g + 4g^2)\|Q - Q'\|_{\max}.$$

Integrating in t proves the assertion, with the displayed slightly larger constant. One may initially replace both covariances by $Q + \epsilon I$ and $Q' + \epsilon I$. The functions and all derivatives used above are bounded, and the partition functions lie between e^{-g} and e^g, so weak Gaussian convergence lets $\epsilon \downarrow 0$. This also proves the singular case. □

Proposition 5.2 (Finite realization of a bounded-tilt tower)**.** *Fix $L \ge 2$, $K \ge 1$, and $G \ge 1$. Let Q_0 be the covariance of the Gaussian first-layer field $w \cdot \bar{x}$, with $w \sim \mathcal{N}(0, I_{d+1}/d)$, indexed by $x \in \mathcal{X}$. Suppose symmetric matrices $\Gamma_1, \dots, \Gamma_{L-1}$ and a vector $\beta \in \mathbb{R}^{\mathcal{X}}$ satisfy*

$$\|\Gamma_s\|_{1,\mathrm{ent}} \le G, \qquad \|\beta\|_1 \le 4K, \qquad Q_s = \mathsf{F}_{\Gamma_s}(Q_{s-1}).$$

Then the function $z(x) = (Q_{L-1}\beta)_x$ can be uniformly approximated to accuracy $1/8$ by a depth-L network whose width and weights are bounded by $\mathrm{poly}_L(d, K, G)$. If $f(x)z(x) \ge 7/8$ everywhere, the finite network implements f with margin at least $3/4$.

Proof. Let $b = 8(1 + G)^2$ and

$$\eta = [32K(L-1)b^{L-2}]^{-1}.$$

Choose the common width n to be the least integer satisfying

$$n \ge 2\eta^{-2} \log(8(L-1)\, 2^{2d}).$$

The construction is random only to prove existence. The matrices Γ_s may be indexed by the entire cube; they select the sampling laws, but are not stored or evaluated by the resulting finite network. No efficient sampling or construction algorithm is claimed. At the first layer, independently sample n Gaussian weight vectors $w_i \sim \mathcal{N}(0, I_{d+1}/d)$ tilted by

$$\exp\{h(w_i)^\top \Gamma_1 h(w_i)\}, \qquad h(w_i)_x = \phi(w_i \cdot \bar{x}).$$

Their activation vectors have exactly the law defining $Q_1 = \mathsf{F}_{\Gamma_1}(Q_0)$. At each subsequent hidden layer, conditional on the already sampled previous activation vectors $h_1, \dots, h_n$, independently sample n standard Gaussian row vectors $\xi_j \in \mathbb{R}^n$ tilted by

$$\exp\{\phi(Z_j)^\top \Gamma_s \phi(Z_j)\}, \qquad Z_j(x) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \xi_{ji} h_i(x).$$

Use the actual network weights $\xi_{ji}/\sqrt{n}$. If $\widehat{Q}_{s-1} = n^{-1} \sum_i h_i h_i^\top$ is the preceding empirical activation covariance, then, conditionally, Z_j before tilting is Gaussian with exactly that covariance. Thus the new activation vectors are conditionally independent, with second moment $\mathsf{F}_{\Gamma_s}(\widehat{Q}_{s-1})$.

Every covariance entry is the empirical mean of n variables in $[-1, 1]$. Hoeffding's inequality, a union bound over 2^{2d} entries, and conditioning at successive layers show, with probability at least $3/4$, that

$$\|\widehat{Q}_s - \mathsf{F}_{\Gamma_s}(\widehat{Q}_{s-1})\|_{\max} \le \eta \quad (1 \le s < L),$$

where $\widehat{Q}_0 = Q_0$. Lemma 5.1 then implies

$$\|\widehat{Q}_{L-1} - Q_{L-1}\|_{\max} \le (L-1)b^{L-2}\eta.$$

For each last-layer activation vector h_i, choose the deterministic readout weight $v_i = (\beta \cdot h_i)/n$. The resulting output is $\widehat{z}(x) = (\widehat{Q}_{L-1}\beta)_x$, and therefore

$$\|\widehat{z} - z\|_\infty \le 4K(L-1)b^{L-2}\eta = 1/8.$$

Although β is indexed by the entire input cube, its only role in the finite network is to select these n real readout weights; each satisfies $|v_i| \le 4K/n$. It is not an additional computational input to the network.

It remains to bound the sampled weights without losing polynomial size. For every sampled row, its tilted density relative to its Gaussian reference is at most e^{2G}, since the quadratic tilt lies in $[-G, G]$. Let $D_n = n(d+1) + (L-2)n^2$ be the total number of incoming scalar coordinates, and choose

$$B = \left[4G + 2\log(8\max\{D_n, 1\})\right]^{1/2}.$$

Each raw incoming coordinate has reference variance at most one, so, also conditionally on all preceding layers, its probability of exceeding B in absolute value is at most $2e^{2G-B^2/2}$. A union bound gives probability at most $1/4$ that any incoming coordinate exceeds B. The actual internal weights divide the raw coordinates by $\sqrt{n}$ and so satisfy the same bound. Intersecting this event with the covariance-approximation event leaves probability at least $1/2$. Fix a realization in the intersection. Both n and $\max\{B, 4K/n\}$ are bounded by polynomials in d, K, G at fixed depth. The stated margin follows from $7/8 - 1/8 = 3/4$. □

Lemma 5.3 (Counting finite realizations)**.** *Fix L, width n, and an entrywise bound $B \ge 1$. The Boolean functions having depth-L, width-n implementations with margin at least $3/4$ and weights bounded by B number at most*

$$\exp\{P\log(3 + 16B\Lambda)\}, \qquad P = n(d+1) + (L-2)n^2 + n, \quad \Lambda = (d+Ln)[1+n(B+1)]^{L-1}.$$

Proof. Round every weight to a multiple of $(8\Lambda)^{-1}$. The first hidden layer's coordinatewise error is at most $(d+1)/(8\Lambda)$. Every subsequent hidden layer has error at most $n(B+1)$ times the preceding error plus $n/(8\Lambda)$; the same estimate applies at the linear readout. Induction bounds the final error by $1/8$. Each rounded coordinate has at most $3 + 16B\Lambda$ possible values, so the P-parameter grid has the stated size. Rounding preserves every represented Boolean sign function. □

6 Likely full-margin draws have small descriptions

Let $m = 2^d$ and $r = L - 1$. Index fields and covariance matrices by the m cube inputs. Define

$$Q_0(x, y) = \bar{x} \cdot \bar{y}/d, \qquad Q_{s,N} = \frac{1}{N}\sum_{i=1}^{N} h_i^s (h_i^s)^{\mathsf{T}} \quad (1 \le s \le r),$$

where h_i^s denotes the activation of neuron i evaluated on all inputs. Write M_Q for the law of $\tanh(Z)$ when $Z \sim \mathcal{N}(0, Q)$. Conditional on the preceding layers, the rows $h_1^s, \dots, h_N^s$ are independent with law $\mathsf{M}_{Q_{s-1,N}}$, with $Q_{0,N} = Q_0$. This is exact: an internal Gaussian row has preactivation covariance $N^{-1}\sum_i h_i^{s-1}(h_i^{s-1})^{\mathsf{T}}$. No covariance matrix is assumed nonsingular.

For a covariance $\widehat{Q}$ and a probability law μ on $[-1,1]^m$, define

$$\mathcal{J}_G(\widehat{Q};\mu) = \max_{\|\Gamma\|_{1,\mathrm{ent}}\le G} \left\{ \langle \Gamma, \widehat{Q}\rangle - \log \int e^{h^\mathsf{T}\Gamma h}\, d\mu(h) \right\}, \quad \langle A, B\rangle = \sum_{i,j} A_{ij}B_{ij}.$$

This is the bounded-observable dual from Section 4 with $D = m^2$. Symmetrizing a maximizing Γ leaves its quadratic form unchanged and cannot increase its entrywise norm.

Fix $K \ge 4$. Choose the layer budgets backward:

$$G_r = 32rK^2, \qquad G_s = 8G_{s+1}(1+G_{s+1})^2 \quad (s = r-1,\dots,1), \qquad b_s = 8(1+G_s)^2. \tag{5}$$

Then $G_s = 32rK^2 \prod_{j=s+1}^{r} b_j$ and $G_1 \ge \cdots \ge G_r \ge 1$. Set

$$q_s = \lceil G_s^2/2 \rceil, \qquad \ell = \log(2m^2+1), \qquad T_K = \left\lceil \max\left\{1, \frac{2q_1\ell}{K-1}\right\} \right\rceil. \tag{6}$$

At fixed depth, G_1, q_1, and T_K are polynomially bounded in d, K; indeed $\ell \le (2d+2)\log 2$ and the cubic recursion gives

$$G_1 = O_L(K^{2\cdot 3^{L-2}}), \qquad T_K = O_L(dK^{4\cdot 3^{L-2}-1}).$$

Lemma 6.1 (A likely set of parameter draws)**.** *Let $\mathcal{E}_K$ be the event that*

$$\mathcal{J}_{G_s}(Q_{s,N}; \mathsf{M}_{Q_{s-1,N}}) \le K \quad \textit{for all } s, \qquad N\|W^L\|_2^2 \le K.$$

For every $N \ge T_K$,

$$\mu_N(\mathcal{E}_K^c) \le Le^{-NK/8}. \tag{7}$$

Proof. Apply Proposition 4.3 conditionally on the preceding layers, taking $s = q_s$ in that proposition. Since $G_s^2/(2q_s) \le 1$ and $q_s \le q_1$, each hidden-layer failure probability is at most

$$\exp\{q_1\ell - N(K-1)\} \le e^{-N(K-1)/2} \le e^{-NK/4}.$$

The bound is uniform in the preceding covariance, so averaging preserves it. For the readout, $N^2\|W^L\|_2^2$ has a chi-square distribution with N degrees of freedom. Its exponential moment at $1/4$ gives

$$\mathbb{P}\{N\|W^L\|_2^2 > K\} \le e^{-NK/4}2^{N/2} \le e^{-NK/8},$$

where $K \ge 4$ is sufficient. A union bound over the r hidden layers and the readout proves the claim. □

Lemma 6.2 (Minimum readout and a bounded dual vector)**.** *Suppose H is any $m \times N$ feature matrix, $g \in \{-1,1\}^m$, and $v \in \mathbb{R}^N$ satisfies $g_i(Hv)_i > 1/2$ for every i and $N\|v\|_2^2 \le K$. Put $Q = HH^\mathsf{T}/N$. There exists $\beta \in \mathbb{R}^m$ with*

$$\|\beta\|_1 \le 2K, \qquad g_i(Q\beta)_i \ge 1/2 \quad \textit{for every } i.$$

This holds even when Q is singular.

Proof. Minimize $N\|u\|_2^2/2$ subject to $g_i(Hu)_i \ge 1/2$. A minimizer exists by coercivity and closedness, and the given v is a strictly feasible point. The elementary convex quadratic-program optimality conditions therefore give multipliers $\lambda_i \ge 0$ with

$$Nu = H^\mathsf{T}\beta, \qquad \beta_i = g_i\lambda_i, \qquad \lambda_i[g_i(Hu)_i - 1/2] = 0.$$

Consequently $Hu = Q\beta$ and

$$\frac{1}{2}\|\beta\|_1 = \beta^\mathsf{T}Hu = N\|u\|_2^2 \le N\|v\|_2^2 \le K.$$

This proves the result without inverting Q. □

Proposition 6.3 (Finite descriptions on the likely set)**.** *For each $K \geq 4$ there is a finite class $\mathcal{H}_K$ of Boolean functions, independent of N, such that every full-margin draw in $\mathcal{E}_K$ has its sign function in $\mathcal{H}_K$, and*

$$\log |\mathcal{H}_K| \leq H_L(d, K),$$

where H_L is polynomially bounded at fixed depth.

Proof. Fix a draw in $\mathcal{E}_K$. Lemma 4.1 gives symmetric matrices Γ_s with $\|\Gamma_s\|_{1,\mathrm{ent}} \leq G_s$ and

$$\|Q_{s,N} - \mathsf{F}_{\Gamma_s}(Q_{s-1,N})\|_{\max} \leq K/G_s.$$

Construct a genuine tilted tower from these same matrices by $Q_0^* = Q_0$ and $Q_s^* = \mathsf{F}_{\Gamma_s}(Q_{s-1}^*)$. The matrices may depend on the entire draw; only their uniform norm bounds will be used to count the resulting functions. For $e_s = \|Q_{s,N} - Q_s^*\|_{\max}$, Lemma 5.1 gives

$$e_0 = 0, \qquad e_s \leq K/G_s + b_s e_{s-1}.$$

The backward choice in (5) yields

$$e_r \leq \sum_{s=1}^{r} \frac{K}{G_s} \prod_{j=s+1}^{r} b_j = \frac{1}{32K}. \tag{8}$$

Using a common budget at every layer would not give this estimate: the larger tilt also enlarges the next layer's covariance Lipschitz constant.

If the draw has full-margin sign g, apply Lemma 6.2 to its final hidden feature matrix and original readout. This gives $\|\beta\|_1 \leq 2K$ and $g_i(Q_{r,N}\beta)_i \geq 1/2$. By (8),

$$\|Q_r^*\beta - Q_{r,N}\beta\|_\infty \leq 1/16.$$

Thus $2Q_r^*\beta$ has signed margin at least $7/8$. Its dual vector 2β has ℓ^1 norm at most $4K$, while all its tilt matrices have entrywise norm at most G_1. Proposition 5.2 supplies a finite network within $1/8$ of this output, and hence with signed margin at least $3/4$. Its width and weights have uniform bounds polynomial in d, K, G_1. Since G_1 is polynomial in K at fixed depth, Lemma 5.3 provides the required finite class and count. □

Here is one explicit choice of the count. Write $G = G_1$, $b = 8(1+G)^2$, and

$$\begin{aligned}
\eta &= [32Krb^{r-1}]^{-1}, & n_* &= \left\lceil 2\eta^{-2}\log(8r\,2^{2d})\right\rceil, \\
D_* &= n_*(d+1) + (L-2)n_*^2, & B_* &= \max\{1, \sqrt{4G + 2\log(8\max\{D_*, 1\})}, 4K/n_*\}, \\
P_* &= D_* + n_*, & \Lambda_* &= (d + Ln_*)[1 + n_*(B_*+1)]^{L-1}.
\end{aligned} \tag{9}$$

Then

$$H_L(d, K) = P_* \log(3 + 16B_*\Lambda_*) \tag{10}$$

is valid. It counts all sign functions realized by the common bounded network family, rather than choices of the input-indexed tilt matrices.

Corollary 6.4 (Aggregate function-mass bound)**.** *For every $N \geq T_K$,*

$$\mu_N\left(\bigcup_{g \notin \mathcal{H}_K} A_{N,g}\right) \leq Le^{-NK/8}.$$

Proof. The union in question is contained in $\mathcal{E}_K^c$ by Proposition 6.3. Apply Lemma 6.1. □

7 The posterior guarantee at every width

We finish with elementary posterior estimates, including the widths below the polynomial threshold T_K.

Lemma 7.1 (A finite relevant class)**.** *Let a prior on a finite function class assign positive mass to a target f. Suppose $f \in \mathcal{H}$, $\log|\mathcal{H}| \le H$, and $\pi(\mathcal{H}^c)/\pi(f) \le \tau$. If $n \ge a^{-1}[H + \log(1/\delta)]$, then, with sample probability at least $1-\delta$, its consistency posterior has average error at most $a + \tau$.*

Proof. A fixed function of true error greater than a survives n independent labels with probability at most $(1-a)^n \le e^{-na}$. A union bound excludes all such survivors in $\mathcal{H}$ except on an event of probability at most e^{H-na}. For every sample the posterior denominator is at least $\pi(f)$, so its mass outside $\mathcal{H}$ is at most τ. □

Proof of Theorem 2.1. Let $A = Q_L(d, C)$ from Proposition 3.1, so $A \ge 1$ and

$$\mu_N(A_{N,f}) \ge e^{-AN/2} \quad \text{for every } N \ge C.$$

Set

$$K = 4A + 8\log(6L/\varepsilon), \qquad P_L(d, C, \varepsilon) = \max\{H_L(d, K),\, AT_K/2\}. \tag{11}$$

These are explicit, and P_L is polynomially bounded in $d, C, \log(1/\varepsilon)$ at fixed depth. We show it suffices to replace p_L in (1) by P_L.

First let $N \ge T_K$. Corollary 6.4, the target-mass bound, and cancellation of the global-margin normalization give

$$\frac{\pi_N(\mathcal{H}_K^c)}{\pi_N(f)} \le L\exp\{-N(K/8 - A/2)\} = L\exp\{-N\log(6L/\varepsilon)\} \le \varepsilon/6.$$

Necessarily $f \in \mathcal{H}_K$, since otherwise this ratio would be at least one. Lemma 7.1, with $a = \tau = \varepsilon/6$, gives $R_D(\Pi_N) \le \varepsilon/3$ with probability at least $1-\delta$ as soon as

$$n \ge \frac{6}{\varepsilon}[H_L(d, K) + \log(1/\delta)].$$

Now let $C \le N < T_K$. The target-mass bound and $\mu_N(A_N) \le 1$ give $-\log \pi_N(f) \le AN/2 \le AT_K/2$. Let B_n be the posterior mass on functions of true error greater than $\varepsilon/6$. Using the target as a lower bound on the posterior normalizer yields

$$\mathbb{E}_{S_n} B_n \le \frac{1}{\pi_N(f)} \sum_{g:\mathrm{err}_D(g) > \varepsilon/6} \pi_N(g)(1 - \mathrm{err}_D(g))^n \le \exp\{AT_K/2 - n\varepsilon/6\}.$$

Markov's inequality gives $B_n \le \varepsilon/6$ with probability at least $1-\delta$ if

$$n \ge \frac{6}{\varepsilon}[AT_K/2 + \log(6/(\varepsilon\delta))].$$

The remaining posterior mass has error at most $\varepsilon/6$, proving $R_D(\Pi_N) \le \varepsilon/3$ in this width range as well. Both sufficient sample bounds follow from (1) with P_L in place of p_L. A polynomial majorant gives the stated theorem.

Finally, at every input whose correct-label posterior probability is at most $2/3$, the posterior error probability is at least $1/3$. Consequently the probability of this failure event is at most $3R_D(\Pi_N) \le \varepsilon$. This proves the strict $> 2/3$ success criterion without excluding equality from the failure event. □

Remark 7.2 (Activation assumptions). Tanh is the activation used in the theorem and throughout the model. The proof also applies to any bounded twice continuously differentiable activation satisfying $|\phi| \le 1$, $|\phi'| \le 1$, and $|\phi''| \le 2$, provided the target has the stated finite bounded-weight implementation. The entropy tests use only bounded features; the covariance comparison uses precisely these derivative bounds. Analyticity, invertibility, and oddness are not used. Other finite derivative bounds change the constants.

The proof establishes a width-uniform statistical guarantee with explicit, though large, polynomial bounds at fixed depth. In particular, it does not assert useful numerical sample sizes or uniform polynomial degrees as depth grows. The finite-width entropy certificates and the backward layer budgets are what permit the same theorem to cover every $N \ge C_L(f)$.